

# Partham Kumar

Karachi, Pakistan

parthamchawla1@gmail.com — +92 336 3428830 — linkedin.com/in/partham-chawla-a8494021a — GitHub

## Summary

Artificial Intelligence graduate with strong foundations in **Machine Learning**, **Deep Learning**, and **Large Language Models**. Experienced in developing AI-driven applications and full-stack systems using **Python**, **FastAPI**, **React**, and **SQL**, with practical experience in embeddings, multi-agent LLM workflows, structured data pipelines, and **REST API development**.

## Education

### Bachelor of Science in Artificial Intelligence

Aug 2022 – June 2026

FAST National University of Computer and Emerging Sciences (NUCES), Karachi

## Technical Skills

|                 |                                                                                   |
|-----------------|-----------------------------------------------------------------------------------|
| Languages       | Python, JavaScript, TypeScript, C/C++, Java, SQL, HTML/CSS, JSON                  |
| AI/ML           | Machine Learning, Deep Learning, Transformers, LLMs, RAG, Embeddings, Fine-tuning |
| Frameworks      | PyTorch, TensorFlow, Scikit-learn, LangChain, LangGraph                           |
| Backend         | FastAPI, Node.js, Express, REST APIs                                              |
| Frontend        | React, Redux Toolkit, Vite, Tailwind CSS                                          |
| Databases/Tools | PostgreSQL, MySQL, MongoDB, Git, Docker, CI/CD, Postman                           |

## Experience

### Freelance Full-Stack Developer & AI Engineer

2023 – Present

- Developed and deployed full-stack applications using React, Node.js, PostgreSQL, MySQL, and MongoDB.
- Designed REST APIs, secure authentication systems, structured data pipelines.

## Projects

### ShopBuddy — XAI Shopping Assistant (Multi-Agent LLM App)

GitHub: [ParthamKumar/ShopBuddy-XAI-Shopping-Assistant](#)

- Built a multi-agent LLM workflow for query parsing, marketplace collection, product ranking, review summarization, and explainable recommendations.
- Implemented web scrapers processing **100+ products** from Amazon and Daraz, with structured JSON outputs and currency conversion.
- Developed a FastAPI backend and React frontend, storing processed product data in PostgreSQL and serving results through REST APIs.
- Added input validation, relevance filtering, and duplicate removal to improve recommendation quality and reliability.

### GPT-Style Language Model & Training Pipeline

- Implemented a GPT-style Transformer from scratch with token embeddings, positional encodings, multi-head self-attention, and autoregressive decoding.
- Built an end-to-end training pipeline using AdamW optimization, learning-rate scheduling, and modular model configuration.
- Implemented greedy decoding, temperature scaling, and top-k sampling for controlled text generation.

### MedVerses — Medical Test Preparation Platform

Live: [www.medverses.com](#)

- Built and deployed a medical QBank platform using React, Redux Toolkit, Node.js, Express, and MySQL.
- Developed custom tests, timed exams, answer tracking, flagged questions, explanations, and performance analytics.
- Implemented JWT authentication, bcrypt password hashing, OTP verification, and role-based access control.
- Created an admin dashboard for managing users, subscriptions, payments, analytics, and account status.
- Added SQL migrations, database backups, Dockerized infrastructure, and GitHub Actions CI/CD.

## Achievements

- Selected for **DevExpo 2024 (University Open House)** as a top finalist for project functionality and design.